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Nutrient concentration and acid–base status of leaf litter of tree species characteristic of the hardwood forest of southern Quebec

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Received January 12, 1992
Accepted June 7, 1993

CÔTÉ, B., and FYLES, J.W. 1994. Nutrient concentration and acid–base status of leaf litter of tree species characteristic of the hardwood forest of southern Quebec. Can. J. For. Res. **24**: 192–196.

Leaf litter of 15 tree species characteristic of the deciduous and mixed forest of southern Quebec were analyzed for pH, directly titrable acids and bases in water extracts, ash bases, excess bases, excess ash bases, and for levels of N, P, K, Ca, and Mg. We hypothesized that many tree species typical of the climax of sugar maple (*Acer saccharum* Marsh.) dominated forest have leaf litter with a higher base status than sugar maple and red maple (*Acer rubrum* L.) leaf litter, and that the base status of leaf litter would be lower on wet sites. Mean differences among species were highly significant ($p < 0.0001$) for all variables related to acidity or bases, but the effect of drainage was not. Red and sugar maple leaf litter was very acid and low in N concentration. American beech (*Fagus grandifolia* Ehrh.) and red oak (*Quercus rubra* L.) leaf litter was not very acidic but was low in nutrient concentrations. White pine (*Pinus strobus* L.) was lowest in all nutrients and ash bases but was low in titrable acidity. Directly titrable bases in leaf litter extracts were correlated positively with leaf litter N and Mg, and ash bases were positively correlated with leaf litter Ca and Mg. Many species typical of the sugar maple climax may have better soil ameliorating potential than sugar and red maple.

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Les litières foliaires de 15 espèces d'arbre caractéristiques de la forêt feuillue et mixte du sud du Québec ont été analysées pour le pH, les acides et les bases directement titrables dans les extraits aqueux de litière, les bases des cendres, l'excès de bases, l'excès de bases dans les cendres et pour les taux de N, P, K, Ca et Mg. Nous avons émis l'hypothèse que plusieurs espèces typiques de la forêt climacique dominée par l'érable à sucre (*Acer saccharum* Marsh.) ont des litières foliaires plus riches en bases que la litière de l'érable à sucre et de l'érable rouge (*Acer rubrum* L.) et que le statut en bases serait plus bas sur stations humides. Les différences entre espèces étaient très significatives ($p < 0.0001$) pour toutes les variables reliées à l'acidité ou aux bases mais l'effet du drainage n'était pas significatif. La litière foliaire de l'érable rouge et à sucre était très acide et pauvre en N. Les litières foliaires du hêtre américain (*Fagus grandifolia* Ehrh.) et du chêne rouge (*Quercus rubra* L.) n'étaient pas très acides mais étaient pauvres en éléments nutritifs. Le pin blanc (*Pinus strobus* L.) avait la litière foliaire la plus pauvre en éléments nutritifs et en bases des cendres mais quoique pas très acide. Les bases titrables dans les extraits de litière foliaire étaient corrélées positivement avec N et Mg, et les bases des cendres étaient corréles positivement avec Ca et Mg. Plusieurs espèces caractéristiques du climax de l'érable à sucre pourraient avoir un meilleur potentiel d'amélioration des sols que l'érable à sucre ou l'érable rouge.

Introduction

The concept that some tree species may ameliorate a forest site has long been established in the forestry literature (Plice 1934; Miles 1985; Howard and Howard 1990). European forest researchers discovered quite early the strong relationships existing between the acidity of leaf litter and its rate of decomposition (Broadfoot and Pierre 1939; Mattson and Koutler-Andersson 1941). Most conifers, through their leaf litters, have been considered to decrease soil fertility by producing acidic mor humus (Messenger 1975; Brand et al. 1986). Complex interactions with soil fertility (Staaf 1987), soil moisture (Taylor and Parkinson 1988) and forest species composition (Blair et al. 1990; Gustafson 1943) do, however, exist and make predictions of rates of litter decomposition very difficult. Recently, Howard and Howard (1990) have rejuvenated the concept of leaf litter acidity by using the concept of excess bases in litter. Excess bases of leaf litter extracts correlate better with the rate of litter decomposition and the type of humus formed than total acidity of litter (Howard and Howard 1990). The concept of excess bases in leaf litter appears to be a simple integrating concept

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of the chemical composition of leaf litter capable of explaining much of the interspecific variation in leaf litter dynamics.

Leaf litters of most important European tree species have been characterized for their chemistry and rate of decomposition (Mattson and Koutler-Andersson 1954) but such synthesis of leaf litter characteristics is not available for tree species of northeastern North America. Among species of the hardwood forest of eastern Canada, eastern hemlock (*Tsuga canadensis* (L). Carr.) and American beech (*Fagus grandifolia* Ehrh.) are well known for their acidifying potential (Ovington 1953; Burns and Honkala 1990). Little is known, however, about the litter of sugar maple (*Acer saccharum* Marsh.) and associated species of the climax forest of southern Quebec. Some characteristics of litters that affect their rate of decomposition were addressed (Melillo et al. 1982; Blair 1988) without providing information that could have forced the revision of the common belief that most North American hardwoods are ameliorating species that produce litter of low acidity. Large differences in the acidity of leaf homogenates were however observed among more than 20 species of hardwoods and conifers of eastern Canada (Schultz and Lechowicz 1986). Sugar maple leaf homogenates (Schultz and Lechowicz 1986) and leaf leachates (Coldwell and DeLong 1950; Liu and Côté 1993) were found to be very acidic as well as having a low buffering capacity (Liu and Côté 1993) compared with other deciduous tree species. Large differences in ameliorating potential may, therefore, exist among deciduous hardwood tree species.

In this study, we hypothesized that many tree species typical of the climax of sugar maple as defined by Grandtner (1966) have leaf litters with a higher base status than sugar and red maple (*Acer rubrum* L.). Our objectives were to characterize the chemistry of leaf litter of tree species of the deciduous and mixed forest of southern Quebec and to determine the effect of soil drainage on directly titrable acids and bases of leaf litter extracts and of ash bases of leaf litter.

Site description

The study site was located in the Morgan Arboretum of McGill University on the West Island of Montreal in southern Quebec (45°25′N, 73°57′W; 30 m above sea level). Most of the arboretum is composed of natural forest stands that range from pioneer to climax forests typical of the Great Lakes – Saint Lawrence forest (Rowe 1972). Soils were derived from fluvial and marine sediments and ranged from Dystric Brunisols to Sombric Brunisols. The pH of the first 10 cm of soil (1:2 of soil and water), exclusive of litter, ranged from 5.5 to 7.5. Stands that were used for the study were of natural origin and were between 50 and 120 years old. Tree species under study were basswood (*Tilia americana* L.), American beech, butternut (*Juglans cinerea* L.), bitternut hickory (*Carya cordiformis* (Wang.) K. Koch.), eastern cottonwood (*Populus deltoides* Marsh.), largetooth aspen (*Populus grandidentata* Michx.), paper birch (*Betula papyrifera* Marsh.), red maple, red oak (*Quercus rubra* L.), shagbark hickory (*Carya ovata* (Mill.) K. Koch.), sugar maple, white ash (*Fraxinus americana* L.), white oak (*Quercus alba* L.), white pine (*Pinus strobus* L.), and yellow birch (*Betula alleghaniensis* Britton).

Materials and methods

Leaf litter sampling and analysis

For six of the 15 tree species under study, six trees were selected in different stands on moist and wet soils (N = 12). For the other nine species, six trees were selected on moist or wet soils (N = 6). Leaf senescence for the different species was monitored closely to synchronize leaf litter sampling with the peak of leaf fall for individual species. Sampling was done over a 3-week period. Falling and recently fallen leaves were collected from the ground around study trees. Leaves were dried at 65°C for 48 h and then ground to pass through a 40-mesh screen. Leaf tissues were digested according to the procedure of Thomas et al. (1967), and analyzed for N and P on a Technicon AutoAnalyzer, and for K, Ca, and Mg by atomic absorption spectrometry. Leaves were analyzed for ash bases and leaf homogenates were analyzed for pH, and directly titrable acids and bases according to the procedure of Howard and Howard (1990) in which solutions were titrated or back titrated to pH 7. Excess bases was defined as the difference between directly titrable bases and directly titrable acids, and excess ash bases as the difference between ash bases and directly titrable acids (Howard and Howard 1990).

Statistical analysis

A completely randomized factorial design was used to test the effect of soil drainage and the interaction between tree species and soil drainage on leaf litter extracts pH, directly titrable acids and bases and excess bases, ash bases, and excess ash bases. For this ANOVA analysis, a data set made of observations on species that could be found on the two soil drainage classes (basswood, American beech, butternut, paper birch, red maple, and sugar maple) was used. The effect of species was tested by ANOVA analyses of the full data set including all species. Partial correlation coefficients were used to assess relationships between acid/base characteristics of leaf litter and leaf litter nutrient concentrations. Statistics were performed using the Statistical Analysis System (SAS Institute Inc. 1985) at a probability level of 5%.

Results

Mean differences among species were highly significant ($p < 0.0001$) for all variables related to acidity or bases but the effect of drainage was not. The interaction between tree species and drainage was significant for the pH of leaf litter extracts only. This interaction resulted from the higher leaf extract pH of paper birch on the moist sites rather than on the wet sites as observed for the other species. The pH of leaf litter extracts was relatively high for basswood, butternut, eastern cottonwood, largetooth aspen, and white ash, and relatively low for red maple, sugar maple, and white pine (Fig. 1). Directly titrable acids were very high for red and sugar maple and intermediate for bitternut hickory, paper birch, and yellow birch (Fig. 1). Species with high leaf litter extract pH also had high directly titrable bases and excess bases. Shagbark hickory had high directly titrable bases and excess bases but average pH of leaf litter homogenates (Fig. 1). Ash bases and excess ash bases were uniformly high for basswood, bitternut hickory, butternut, eastern cottonwood, largetooth aspen, paper birch, shagbark hickory, white ash, and yellow birch, and relatively low for American beech, red and white oak, red maple, and white pine (Fig. 1). Sugar maple was high in ash bases and relatively low for excess ash bases (Fig. 1).

Based on highest and lowest quartiles, basswood, butternut and bitternut hickory leaf litters were consistently high in all nutrients except K in butternut and P in bitternut hickory (Table 1). Leaf litter of eastern cottonwood was high in N, K, and Ca; white ash, in P and K; yellow birch, in P and Mg; and shagbark hickory, in Mg. Red maple, American beech, red oak, and white pine leaf litters were consistently low in all nutrients. Leaf litter of white oak was low in K and Mg and sugar maple, in N.

Correlation coefficients between leaf litter extract chemical characteristics, ash bases, and leaf litter nutrient concentrations revealed that (*i*) directly titrable bases in leaf

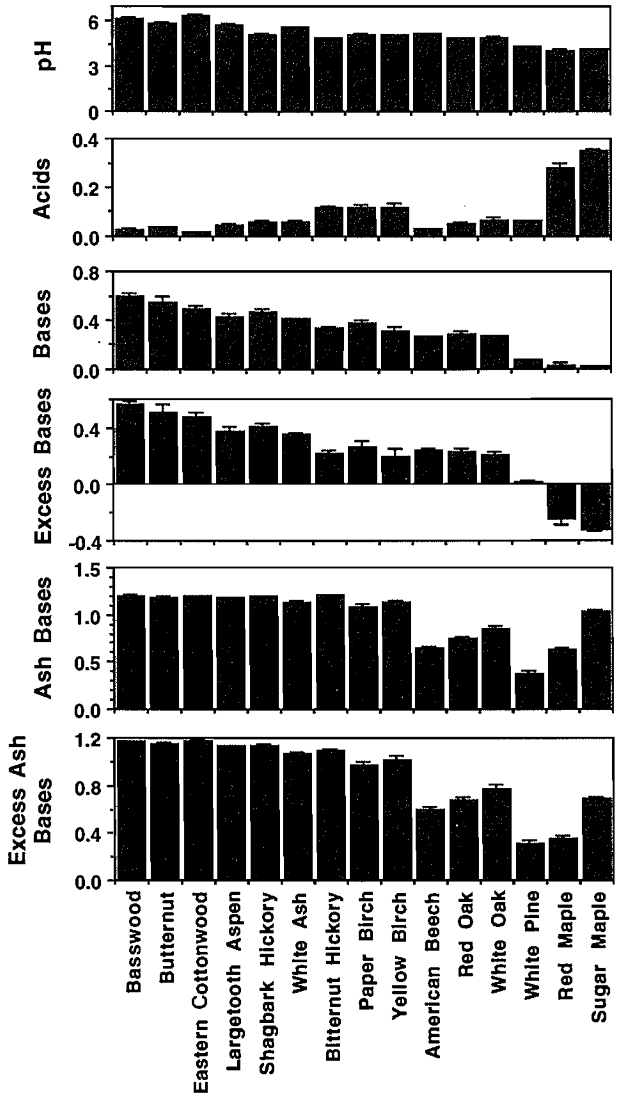

FIG. 1. Mean leaf litter extract pH, directly titrable acids and bases and excess bases, ash bases, and excess ash bases in leaf litter of tree species characteristic of the hardwood forest of southern Quebec. Error bars indicate 1 SE.

litter extracts were correlated positively with leaf litter N and Mg and negatively correlated with leaf litter K; (*ii*) ash bases were positively correlated with leaf litter Ca and Mg; (*iii*) directly titrable acids were negatively correlated with leaf litter N; and (*iv*) the pH of leaf litter extracts was positively correlated with N, P, and Mg (Table 2).

Discussion

Our results for directly titrable acids in paper and yellow birch, American beech, and white ash were consistent with the values observed by Howard and Howard (1990) for *Betula pendula* × *pubescens, Fagus sylvatica* L., and *Fraxinus excelsior* L., respectively. Directly titrable acids in leaf litter extracts of red and sugar maple in our study were twice the highest value observed by Howard and Howard (1990) for directly titrable acids in western hemlock (*Tsuga heterophylla* (Raf.) Sarg.). As a group, red and sugar maple have very acidic leaf litter extracts. Directly titrable bases were also in general agreement with the values observed by Howard and Howard (1990) although minimum and maximun values observed in their study for western hemlock and elm (*Ulmus glabra* Hudson), respectively, were not matched by any of the species at our site. In both studies, the ranking of species for directly titrable bases and acids generally followed a reverse order.

Ash bases were relatively high in our study. Nine species had levels equivalent to the highest concentration observed by Howard and Howard (1990) for elm. Although American beech, red oak, and red maple had relatively low levels of ash bases in our study, their levels were nevertheless higher than for the coniferous species studied by Howard and Howard (1990). Ash bases are mainly derived from cations in insoluble precipitates such as calcium oxalate or in metal–organic compounds, which do not dialyse (Mattson and Koutler-Andersson 1945). Our results confirmed that ash bases in leaf litter are related to Ca and Mg compounds. The relatively high ash bases observed in our study might be related to the high Ca and Mg status of the soils found in the Morgan Arboretum (Liu and Côté 1993).

Site was shown to significantly affect directly titrable acids and bases in leaf litter extracts of many species (Howard and Howard 1990). In our study, the effect of drainage on these variables was generally not significant except for one species, paper birch. The difference may have been that our sites differed markedly in drainage but not in fertility. The significant effect of drainage on the pH of leaf litter extract of paper birch was likely to be too small to have any ecological significance.

The combined analysis of acid–base characteristics and nutrient concentrations of leaf litter provides a basis for grouping of tree species with potentially similar effects on leaf litter dynamics. For one, red and sugar maple leaf litter is characterized by very high acidity and low N concentrations, two factors known to affect litter decomposition and nutrient turnover (Melillo et al. 1982; Berg 1986; Blair 1988). Red maple was also low in P, K, Ca, and Mg. American beech and red oak leaf litter is not very acidic but is low in nutrient concentrations. Such low nutrient concentrations may be related to the high lignin content of their leaves (Coldwell and DeLong 1950), a characteristic that could affect their palatability, and, therefore, their rate of decomposition. Most other species of hardwoods had leaf litter that were markedly higher in titrable bases, ash bases and nutrient concentrations and lower in acidity and, therefore, should have ameliorating properties. Some of the accompanying species of sugar maple in southern Quebec such as *Ulmus americana* L., white ash and basswood, are already known to have leaf litters of low acidity that decompose rapidly (Wittich 1943; Boudru 1947). Birch species have long been recognized for their site ameliorating effects, which include the capacity to reduce soil acidity and increase Ca availability (cf. Miller 1984). The results suggest that many species typical of the sugar maple climax may have better soil ameliorating potential than sugar and red maple.

Sugar maple stands are commonly found on moderately acidic soils with low cation exchange capacity, which could be easily acidified (Johnson et al. 1982). Under present conditions of acid deposition and forest management, most forests of eastern North America may not be self-sufficient in key nutrients such as Ca (Federer et al. 1989). Maple bushes are managed differently than forest stands used for wood production. High grading and (or) selection thinning have been used extensively for decades in maple bushes of southern Quebec to favor sugar maple and maximize the production of maple syrup. Most sugar maple stands of

TABLE 1. Leaf litter nutrient concentrations of tree species characteristic of the hardwood forest of southern Quebec

Species	Nutrient concentrations ($mg \cdot g^{-1}$) N	P	K	Ca	Mg
Basswood	11.2±0.4	1.3±0.08	7.7±0.8	33.8±0.9	4.8±0.09
Butternut	12.1±0.6	1.5±0.3	4.7±0.3	36.4±1.8	6.2±0.5
Eastern cottonwood	11.0±0.8	0.96±0.08	7.2±1.1	29.5±1.7	4.4±0.4
Largetooth aspen	9.3±1.0	0.95±0.05	5.7±0.4	27.5±1.4	3.5±0.2
Shagbark hickory	8.6±0.3	0.91±0.05	4.5±0.7	25.9±0.5	4.8±0.2
White ash	8.6±0.7	1.6±0.2	9.6±0.1	22.5±1.1	3.3±0.2
Bitternut hickory	10.8±0.5	1.1±0.04	6.1±0.6	28.5±1.4	4.5±0.1
Paper birch	9.3±0.4	0.92±0.1	5.1±0.7	17.0±1.2	4.4±0.3
Yellow birch	7.8±0.2	1.2±0.3	3.1±0.5	18.7±1.8	5.0±0.3
American beech	7.2±0.2	0.43±0.08	1.4±0.2	9.6±0.3	2.4±0.1
Red oak	7.2±0.2	0.78±0.02	2.4±0.1	11.6±0.4	2.4±0.03
White oak	9.6±0.2	1.2±0.06	2.3±0.2	15.1±0.8	2.2±0.1
White pine	5.0±0.1	0.31±0.02	0.54±0.1	6.3±0.5	1.2±0.06
Red maple	6.5±0.5	0.60±0.1	2.4±0.5	9.8±0.4	2.1±0.04
Sugar maple	6.6±0.3	1.1±0.1	3.8±0.3	18.5±0.4	2.7±0.07

NOTE: Values are means ±1 SE.

TABLE 2. Partial coefficients of correlation between acidity–basicity characteristics of leaf litter and leaf litter nutrient concentrations of the 15 tree species with associated level of probability given in parentheses (N = 124)

	pH	Directly titrable acids	Directly titrable bases	Ash bases	N	P	K	Ca	Mg
pH		−0.63 (0.0001)	0.69 (0.0001)	0.21 (0.03)	0.34 (0.0003)	0.30 (0.001)	0.04 (0.72)	0.16 (0.10)	0.25 (0.008)
Directly titrable acids			−0.73 (0.0001)	−0.15 (0.12)	−0.42 (0.0001)	−0.06 (0.55)	0.12 (0.21)	−0.03 (0.74)	−0.14 (0.15)
Directly titrable bases				0.27 (0.004)	0.24 (0.01)	−0.04 (0.68)	−0.23 (0.01)	0.09 (0.33)	0.37 (0.0001)
Ash bases					0.04 (0.71)	−0.13 (0.19)	0.11 (0.24)	0.56 (0.0001)	0.39 (0.0001)

southern Quebec have been transformed from a forest composed of many deciduous species (maples, ashes, elms, hickories, basswood, beech, etc.) to almost a monoculture of old sugar and (or) red maple. Increases in leaf litter acidity may have resulted in increasing soil acidity in sugar maple and especially red maple stands subjected to selective cuttings. Recent evidence of the acidifying potential of maple was provided by the study of France et al. (1989) in which the increase in forest floor acidity over a 27-year period was found to be maximum for maple compared with white pine, white spruce (*Picea glauca* (Moench) Voss) and paper birch. The gradual transformation of mixed to pure stands of sugar and red maple in eastern Canada may have resulted in litter of increasing acidity and soil acidification with an accompanying decrease in base cation availability and soil fertility. Widespread decline of sugar maple forests in eastern Canada could have been exacerbated by conversion of mixed deciduous forests to predominantly maple forests.

White pine, the only conifer included in the study, was lowest in all nutrients and ash bases but was low in titrable acidity. Low uptake of nutrients by conifers has been associated with slower rate of soil acidification compared to some nutrient demanding hardwoods (Alban 1982; Johnson and Todd 1990). Although important, leaf litter characteristics alone can not explain the evolution of forest soil acidity nor some of the interspecific and intraspecific differences in the rates of litter decomposition. For example, leaf litter of European beech is neither strongly acid nor does it have low excess bases (Howard and Howard 1990) but it decomposes slowly (Heath and Arnold 1966). Other chemical or physical characteristics of the substrates have to be taken into consideration to understand such exceptions. Variables such as leaf litter nutrients, C:N, lignin, and free amino acids have been used with a certain degree of success to explain the rate of litter mass loss during the different phases of litter decomposition (Berg 1986). Other chemical and physical variables known to affect the palatability of leaves such as tannins and toughness (Coley 1987) will have to be integrated to predict the rate of litter decomposition.

In order to explore the full potential contribution of acid–base status of leaf litter to the understanding of the dynamics of leaf litter and pedological processes in forest ecosystems, we will have to identify and quantify the substances contributing to acidity, directly titrable bases, ash bases and of all nutrients in litter. Some of these substances have been characterized in a few tree species (Blaser et al. 1984; Brown and Sposito 1991) and application to ecosystem processes is promising. The contribution of roots to nutrient cycling and soil fertility will also have to be addressed because of the important input of nutrients and carbon through that pathway (McClaugherty et al. 1982; Raich and Nadelhoffer 1989). Further studies on the effect of each species on soil characteristics when grown on different types of soils and in mixture with different species will provide useful information for understanding and maintaining the sustainability of forest ecosystems.

Acknowledgements

Funding was provided by the Natural Sciences Engineering Research Council of Canada (Grant no. OGP0041725). The significant contribution of Stephen Yamasaki and Juan Li to this study is gratefully acknowledged.

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